RESEARCH ARTICLE

Comparing Acute and 1-Year Outcomes Between Fall- and Motor Vehicle–Related Traumatic Brain Injury

A NIDILRR TBI Model Systems Study

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Neurology® 2026;106:e214775. doi:10.1212/WNL.0000000000214775

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Abstract

Background and Objectives

Traumatic brain injury (TBI) mechanisms are often grouped together in research. Differences in acute and long-term outcomes across mechanisms of injury (MOIs) remain unclear, partly because of confounding by age. Modeling MOI-specific effects can inform clinical triage and prognostication. We examined the relationship between motor vehicle accidents (MVAs) vs falls, the 2 most common MOIs, and acute and 1-year post-injury outcomes, after rigorous control of demographic and preinjury personal factors.

Methods

Data were analyzed from individuals with moderate-to-severe TBI requiring inpatient rehabilitation from the TBI Model Systems National Database, a multicenter prospective longitudinal cohort study. The analytic sample was restricted to individuals aged 16–79 years with an MOI due to MVA or fall occurring between April 2010 and January 2023. We used inverse probability of treatment weighting, based on propensity scores, to adjust for 14 demographic and preinjury personal characteristics and estimate the causal effect of MOI on acute and 1-year outcomes after TBI. Acute hospital and rehabilitation outcomes included the following: Glasgow Coma Scale (GCS), sedation, intubation, post-traumatic amnesia duration, time to follow commands (TFC), length of hospital stay (LOS), and Functional Independence Measure (FIM) cognitive and motor scores. One-year outcomes included the following: Disability Rating Scale and Participation Assessment with Recombined Tools Objective.

Results

Among 5,181 participants (mean age 45.1 ± 19.5, 70% male), 48.4% sustained their injury from MVAs and 51.6% from falls. After weighting and multiple comparisons adjustment, the MVA group had lower GCS total scores by 1.27 points (95% CI −1.92 to −0.61; adjusted $p = 0.001$), greater odds of receiving sedation (odds ratio 1.43, 95% CI 1.11–1.85; adjusted $p = 0.014$), longer TFC by 1.64 days (95% CI 0.39–2.89; adjusted $p = 0.017$), and lower discharge FIM motor scores by 4.28 points (95% CI −7.50 to −1.26; adjusted $p = 0.014$). At 1 year after injury, disability levels and community participation did not differ.

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Glossary

CBI-M = Clinical, Biomarker, Imaging, and Modifier; **DRS** = Disability Rating Scale; **ED** = emergency department; **FIM** = Functional Independence Measure; **GCS** = Glasgow Coma Scale; **ICD** = International Classification of Diseases; **IPTW** = inverse probability of treatment weighting; **KS** = Kolmogorov-Smirnov; **LOS** = length of stay; **MOI** = mechanism of injury; **MVA** = motor vehicle accident; **NDSC** = National Data and Statistical Center; **OR** = odds ratio; **PART-O** = Participation Assessment with Recombined Tools Objective; **PTA** = post-traumatic amnesia; **SMD** = standardized mean difference; **TBI** = traumatic brain injury; **TBIMS** = TBI Model Systems; **TFC** = time to follow commands.

Discussion

MVA-related TBI was associated with worse acute outcomes. However, by 1 year after injury, disability level and community participation do not differ. This work highlights novel findings in short-term and long-term outcomes after falls and MVAs, the leading TBI causes, which are not explained by confounders such as age. Findings may not generalize beyond patients receiving inpatient rehabilitation for TBI.

Introduction

Traumatic brain injury (TBI) is a major cause of disability and mortality across the lifespan.[1] TBI results from various mechanisms of injury (MOIs), including falls, motor vehicle accidents (MVAs), violence, sports injuries, and self-injurious behavior.[2,3] Studies of TBI, including clinical trials,[4] commonly group different MOIs, statistically assuming homogeneity in outcomes across mechanisms. Yet, MOIs may differentially influence short-term and long-term outcomes, as MVAs often involve high-impact velocities and greater kinetic energy transfer relative to other injury types,[5] particularly falls,[6] which may lead to greater acute injury severity and polytrauma. Previous work in severe TBI found that individuals with traffic-related injuries present with greater acute severity than falls but had better 1-year functional outcomes after adjusting for age and injury-severity indicators.[7] However, these models did not account for preinjury sociodemographic and contextual factors.

Systematic group differences exist between individuals who sustain the 2 most common MOIs, MVAs and falls, which together account for over two-thirds of all TBIs.[8] For example, falls are more prevalent among older adults and female individuals,[2,9] whereas MVAs are more prevalent among younger adults and male individuals.[2,10] Chronic medical conditions, such as stroke and diabetes, increase fall risk,[11,12] but less so for MVA. MVA rates are higher in rural areas while more fall-related injuries occur in urban areas.[13] These systemic differences in MOI occurrence also affect outcomes. Demographic and personal preinjury factors such as age,[14-16] race,[17] education,[18] insurance status,[19] psychiatric history,[20] marital status,[21] residence,[22] employment,[23] military service history,[24] rurality,[13] and medical comorbidities[15] are associated with short-term and long-term TBI outcomes. For instance, older age is associated with worse functional outcomes[14] and reduced community participation.[16] These differences complicate statistical comparisons and limit understanding of how outcomes vary by mechanism. Addressing this could affect decisions about pooling TBI mechanisms in clinical trials and longitudinal studies.

To enable robust comparison between those with MVAs and falls, we leveraged data from the TBI Model Systems (TBIMS) National Database to compare acute and 1-year TBI outcomes between individuals who sustained MVA vs fall-related TBI. Building on previous evidence that outcomes vary by mechanism,[7] and to address confounding beyond age and injury-severity markers, we reexamine this question using a weighting approach that accounts for a broader set of preinjury characteristics. We used inverse probability of treatment weighting, based on propensity scores, to rigorously control for known confounders and enable valid comparisons between systematically different injury types. We hypothesized that after controlling for systematic differences in demographic and preinjury personal factors, individuals with MVA-related TBI would demonstrate worse acute and 1-year postinjury outcomes compared with those with fall-related TBI.

Methods

Participants

We used data from the TBIMS National Database, a multicenter prospective longitudinal cohort study of individuals with TBI recruited from inpatient rehabilitation hospitals across the United States. Participants were aged ≥16 years and had a TBI with post-traumatic amnesia (PTA) >24 hours, loss of consciousness >30 minutes, Glasgow Coma Scale (GCS) score <13 on emergency department (ED) admission, or intracranial abnormalities on CT imaging. All participants were admitted to the ED within 72 hours of injury and completed inpatient rehabilitation. Follow-up was conducted through phone or email at 1, 2, 5, and every subsequent 5 years until death or loss to follow-up.

We restricted the sample to individuals with an MOI of MVA or fall, given known differences in exposure risk (e.g., age), distinct injury dynamics related to impact velocity and energy transfer, and that they account for 70% of TBIs in the TBIMS National Database. We limited the sample to individuals aged

16–79 because few ≥80-year-olds sustained a TBI due to MVA (n = 92), aligning with the positivity assumption in causal inference.[25] To minimize missing data, we included participants injured between April 1, 2010, and January 1, 2023, because several confounder variables (e.g., military service and International Classification of Disease [ICD] codes) were not collected before this date. We excluded individuals with missing covariate data (n = 448, 7.9%). Figure 1 presents the study flow diagram. We assessed potential selection bias by comparing the final analytic sample with those excluded because of missing data in eTable 1.

Standard Protocol Approvals, Registrations, and Patient Consents

Participants or their proxy provided informed consent, and the local Institutional Review Board approved study protocols at each site.

Measures

Primary Exposure

MOI (MVA or fall) was identified using the TBIMS mechanism of injury variable, based on data collectors' medical chart review and linked to the primary ICD external cause of injury code.

Covariates

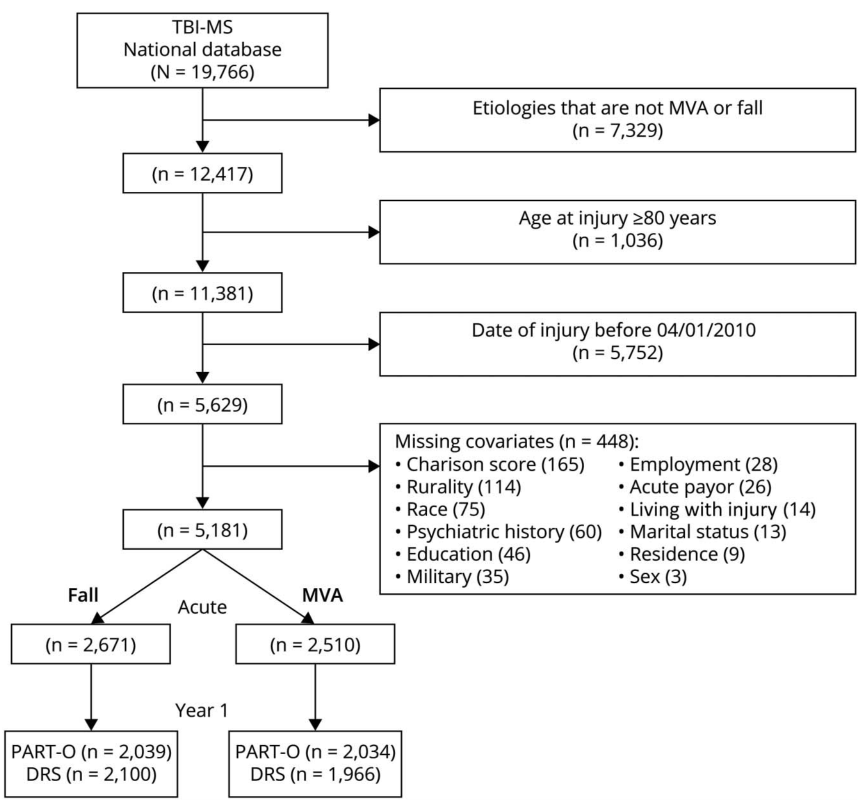
We selected confounders based on known association with MOI and/or injury outcomes of interest, including sex, race, education, primary payor for acute hospitalization, age at injury, preexisting psychiatric history, military service, marital status, cohabitation status, residence and rurality, employment, comorbidities, and orthopaedic injuries at the time of injury determined by ICD codes in the medical record.[26] Variable coding mostly followed the National Data and Statistical Center (NDSC) TBIMS data dictionary,[27] with details in eMethods.

Acute Postinjury Outcomes

Acute care and inpatient rehabilitation outcomes included the following: GCS score (total and subscales), acute care sedation and intubation, duration (days) of PTA, time to follow commands (TFC; days), length of stay (LOS; days) in acute and inpatient rehabilitation, and Functional Independence Measure (FIM) cognitive and motor scores at inpatient rehabilitation admission and discharge.

We used the GCS at ED admission (total and eye opening, verbal, motor subscale scores). Total GCS scores range from 3 to 15 (3–8: severe injury; 9–12: moderate injury; 13–15: mild injury).[28] Patients who were chemically paralyzed, sedated, or intubated were retained in the sample by imputing GCS subscale scores with 1 (i.e., GCS total of 3).[29] GCS eye and motor scores were imputed for 1,178 participants, GCS verbal for 1,925 participants, and GCS total for 1,921 participants.

Figure 1 Study Flow Diagram



DRS = Disability Rating Scale; MVA = motor vehicle accident; PART-O = Participation Assessment with Recombined Tools Objective; TBI-MS = Traumatic Brain Injury Model Systems.

Duration of PTA was established by review of acute care medical records or prospectively tracked during inpatient rehabilitation using the Galveston Orientation and Amnesia Test[30,31] or the Orientation Log.[32] For patients discharged from inpatient rehabilitation while still in PTA (n = 833), duration of PTA was calculated as the hospital LOS (acute and rehabilitation) plus 1 day as per a standard NDSC practice.[29]

TFC was defined as the number of days from injury to 2 consecutive days of simple motor command following. For individuals who never followed commands (n = 82), TFC was calculated as hospital LOS plus 1 day.

Acute hospital LOS was calculated as the number of days between ED admission and acute care discharge. Inpatient rehabilitation LOS was the number of days between inpatient rehabilitation admission and discharge.

The FIM measures independence in daily activities, with a 13-item motor subscale and a 5-item cognitive subscale.[33,34] Each item was scored from 1 (*total assistance*) to 7 (*complete independence*), producing total scores of 13–91 (motor) and 5–35 (cognitive); higher scores indicate better functioning.

One-Year Postinjury Outcomes

The Disability Rating Scale (DRS) measures global disability across 8 areas: verbalization, motor response, eye opening, cognitive capacity for feeding, toileting, grooming, and overall independence and employability.[35] Each is rated on a scale of 0–3 or 5 (total scores 0–30), with higher scores representing greater disability; a score of 30 indicates death.

The Participation Assessment with Recombined Tools Objective (PART-O) measures the frequency of community participation after rehabilitation across 3 domains: (1) productivity, (2) out and about, and (3) social relationships.[36,37] Items are scored from 0 to 5, with higher scores indicating greater participation. We also examined the average total score.

Statistical Analysis

We summarized demographic and clinical variables in the overall sample and by MOI (MVA vs fall) using descriptive statistics. To adjust for confounding by demographics and preinjury personal factors, we applied inverse probability of treatment weighting (IPTW) using propensity scores. The propensity score[38] is a balancing score estimated as the probability of exposure (MVA-related or fall-related TBI) conditional on a set of covariates. We examined interactions between covariates and both age and sex when constructing the propensity model and retained significant interactions. Stabilized inverse probability of treatment weights were calculated from propensity scores estimated through multivariable logistic regression using an average treatment effect estimand.[39] Absolute standardized mean differences (SMDs), variance ratios, and Kolmogorov-Smirnov (KS) statistics for covariates before and after weighting were used as exposure model diagnostics. Good balance was indicated by absolute SMD <0.1, variance ratios near 1, and KS statistics near 0.[40]

We estimated the effect of MVA-related vs fall-related TBI on acute and 1-year injury outcomes using IPTW. Linear regression, weighted by the stabilized IPTW, estimated the effect for continuous outcomes (GCS, PTA duration, TFC days, FIM, acute and rehabilitation LOS, DRS, and PART-O), and weighted logistic regression estimated the odds ratio (OR) for exposure effect for dichotomous outcomes (sedation, intubation). We used robust, sandwich-type, standard errors to not underestimate variance in models with survey weights and to account for the uncertainty in the propensity score estimation. We reported both unweighted and IPTW analyses for all outcomes. For the IPTW analyses, the Benjamini-Hochberg method[41] was used to correct for multiple testing, separately on acute and 1-year outcomes (adjusted p value).

For acute outcomes, we used all available data. Missing data were generally low (around 1%–3%) for most measures but reached up to 17% for FIM motor scores. For the 1-year follow-up, some participants did not have outcome data for reasons such as withdrawal, refusal, or the outcome not being collected. A comparison of those with and without 1-year data is presented in eTables 2 and 3. Given that a reduced sample was used for analysis of 1-year postinjury outcomes, we re-assessed balance on covariates and included any covariates that were no longer well-balanced as covariates in regression models to account for residual confounding. We conducted all analyses using R 4.4.1[42] and used the *WeightIt* package for propensity score weighting.[43]

Data Availability

Data are available through the TBIMS NDSC (tbindsc.org/Researchers.aspx).

Results

The analysis cohort (n = 5,181) had a mean (SD) age of 45.1 (19.5) years, was 70% male and 66.5% non-Hispanic White, and had 13 (3) years of education; 48.4% had sustained their index injury from an MVA (Table 1). Before weighting, age was the most unbalanced factor (SMD = 1.38). After propensity score weighting, the absolute SMDs were <0.1, variance ratios were generally close to 1, and KS statistics were near 0 for all covariates (Table 1; eFigure 1), indicating appropriate balance.[40] Final standardized weights had a mean of 1.0 across groups, with similar medians (0.6) and interquartile ranges (0.5–0.8); weights ranged 0.49–67.5 in the overall group, 0.52–55.9 in the fall group, and 0.49–67.5 in the MVA group.

We examined whether the amount of missing outcome data differed between the MVA and fall groups. Before weighting, the MVA group had a higher likelihood of missingness across several outcome measures. However, after adjusting for covariate differences using propensity scores, differences in missingness rates were no longer present. This suggests that the initial imbalance in missingness was associated with demographic and preinjury personal differences between groups, not with the MOI itself.

Table 1 Sample Characteristics

	Total (N = 5,181)	Fall (N = 2,671)	MVA (N = 2,510)	Absolute mean standardized difference (before IPTW)[a]	Absolute mean standardized difference (after IPTW)[b]
Age at index injury, mean (SD)	45.1 (19.5)	55.8 (16.6)	33.70 (15.4)	1.38	0.02
Sex, n (%)					
Male	3,625 (70.0)	1,953 (73.1)	1,672 (66.6)	0.07	0.05
Female	1,556 (30.0)	718 (26.9)	838 (33.4)		
Race, n (%)					
Non-Hispanic White	3,447 (66.5)	1,898 (71.1)	1,549 (61.7)	0.09	0.01
Non-Hispanic Black	768 (14.8)	290 (10.9)	478 (19.0)	0.08	0.01
Hispanic	702 (13.5)	331 (12.4)	371 (14.8)	0.02	0.01
Other (including Asian/Pacific Islander and Native American)	264 (5.1)	152 (5.7)	112 (4.5)	0.01	0.01
Education, mean (SD)	13.0 (3.0)	13.3 (3.4)	12.6 (2.5)	0.22	0.04
Premorbid psychiatric history, yes, n (%)	1,207 (23.3)	662 (24.8)	545 (21.7)	0.03	0.01
Primary acute payor, n (%)					
Medicare	1,006 (19.4)	890 (33.3)	116 (4.6)	0.29	0.01
Medicaid	959 (18.5)	428 (16.0)	531 (21.2)	0.05	0.01
Private insurance/WC	2,292 (44.2)	1,206 (45.2)	1,086 (43.3)	0.02	0.02
No fault auto	492 (9.5)	16 (0.6)	476 (19.0)	0.18	0.00
Other acute payors	432 (8.3)	131 (4.9)	301 (12.0)	0.07	0.00
Marital status at injury, n (%)					
Single	2,235 (43.1)	707 (26.5)	1,528 (60.9)	0.34	0.00
Married	1,912 (36.9)	1,313 (49.2)	599 (23.9)	0.25	0.02
Other marital status	1,034 (20.0)	651 (24.4)	383 (15.3)	0.09	0.01
Living with injury, n (%)					
Alone	919 (17.7)	556 (20.8)	363 (14.5)	0.06	0.02
Spouse/significant other	2,304 (44.5)	1,456 (54.5)	848 (33.8)	0.21	0.01
Other family	1,601 (30.9)	485 (18.2)	1,116 (44.5)	0.26	0.01
Someone else	357 (6.9)	174 (6.5)	183 (7.3)	0.01	0.00
Residence at injury, n (%)					
Private residence	5,074 (97.9)	2,594 (97.1)	2,480 (98.8)	0.02	0.01
Preinjury employment, n (%)					
Competitively employed/student	3,321 (64.1)	1,343 (50.3)	1,978 (78.8)	0.29	0.01
Retired	766 (14.8)	675 (25.3)	91 (3.6)	0.22	0.00
Other employment	1,094 (21.1)	653 (24.4)	441 (17.6)	0.07	0.00
Military service history, yes, n (%)	597 (11.5)	426 (15.9)	171 (6.8)	0.09	0.02
Rurality of preinjury residence, n (%)					
Rural	1,377 (26.6)	516 (19.3)	861 (34.3)	0.15	0.02
Urban	2,287 (44.1)	1,452 (54.4)	835 (33.3)	0.21	0.05

Continued

Table 1 Sample Characteristics *(continued)*

	Total (N = 5,181)	Fall (N = 2,671)	MVA (N = 2,510)	Absolute mean standardized difference (before IPTW)[a]	Absolute mean standardized difference (after IPTW)[b]
Suburban	1,517 (29.3)	703 (26.3)	814 (32.4)	0.06	0.03
Charlson index score, mean (SD)	0.7 (1.0)	0.9 (1.1)	0.4 (0.7)	0.50	0.04
Orthopedic injury, yes, n (%)	3,020 (58.3)	1,005 (37.6)	2,015 (80.3)	0.43	0.01

Abbreviations: IPTW = inverse probability of treatment weighting; MVA = motor vehicle accident; WC = worker's compensation; TBI = traumatic brain injury.
[a] Calculated by standardizing the difference of the means for continuous variables and proportions for categorical variables in those who sustained a fall-related TBI from those who sustained an MVA-related TBI.
[b] Adjusted for demographic and preinjury covariates using propensity score weighting.

Acute Outcomes

Descriptives of the acute outcomes are summarized in Table 2. Relative to unweighted models, estimates of the weighted models were attenuated but indicated strong associations (Figure 2; eTable 4). Specifically, after IPTW, having an MVA-related vs fall-related TBI was associated with worse GCS subscale scores (Eye −0.30, 95% CI −0.49 to −0.11, adjusted p = 0.006; motor −0.57, 95% CI −0.84 to −0.30, adjusted p = 0.001; verbal −0.40, 95% CI −0.64 to −0.15, adjusted p = 0.006; total −1.27, 95% CI −1.92 to −0.61, adjusted p = 0.001), higher odds of receiving sedation (OR 1.43, 95% CI 1.11–1.85, adjusted p = 0.014), and marginally higher odds of intubation (OR 1.45, 95% CI 1.02–2.04, adjusted p = 0.056). MVA-related TBI was also associated with more days to TFC (1.64, 95% CI 0.39–2.89, adjusted p = 0.017) and worse FIM motor scores at inpatient rehabilitation admission (−3.63, 95% CI −6.31 to −0.96, adjusted p = 0.016) and discharge (−4.38, 95% CI −7.50 to −1.26, adjusted p = 0.014) compared with fall-related TBI.

One-Year Outcomes

One-year outcomes are summarized in Table 2. At 1-year after injury, 4,066 participants (MVA n = 1,966; fall n = 2,100) had DRS data, and 4,343 participants (MVA n = 2,034; fall n = 2,039) had PART-O data available. Participants with 1-year outcome data differed systematically from those without, as well as were more educated and more likely to have private insurance, to live in private residences, and be employed. Those with DRS data were also more likely to be White, have greater social support, and have a military history, while those with PART-O data were generally younger and more likely to live in urban areas (eTables 2 and 3).

Given that the sample with DRS data was reduced relative to the sample used to calculate the propensity scores, we compared covariate balance across all covariates in the reduced sample, which remained balanced (eTable 5). One-year DRS scores between MVA-related and fall-related TBI groups did not differ (0.40, 95% CI −1.49 to 2.30, adjusted p = 0.731; Figure 3A; eTable 6).

In the PART-O sample, education was the only covariate no longer balanced (SMD = −0.14; eTable 7) and was thus further adjusted in the weighted model. Relative to unweighted models, all estimates in the weighted models were attenuated and no longer indicated strong associations. MVA-related TBI was associated with higher PART-O out and about scores (0.13, 95% CI 0.01–0.26, p = 0.03) scores compared with fall-related TBI (Figure 3B; eTable 6) but was no longer significant once adjusted for multiple comparisons (adjusted p = 0.15). There were no clinically important differences by mechanism in other PART-O subscales or total scores.

Discussion

This cohort study of 5,181 participants with moderate-to-severe TBI compared acute and 1-year outcomes between the 2 most common injury types: MVAs and falls. After balancing demographic and preinjury differences, MOI uniquely contributed to both short-term and long-term outcomes. Individuals with MVA-related TBI had greater injury severity and worse acute motor function than those with fall-related TBI. However, by 1-year after injury, disability levels were not different, and individuals with MVA-related TBI showed modestly greater community participation outside the home, although this finding was not robust after correction for multiple comparisons. These findings partially support our hypothesis, showing worse acute outcomes for MVA-related TBI but no sustained differences at 1 year. By rigorously accounting for confounders, we demonstrated the direct contribution of MOI (MVA vs fall) on clinical and functional outcomes, which may inform triage, prognostication, and future research on mechanism-specific rehabilitation.

Shortly after injury, participants with MVA-related TBI showed worse clinical and functional outcomes, even after balancing preinjury demographic and personal factors, suggesting that the higher velocity nature of MVAs may contribute to greater acute severity.[7] Although the fundamental components of the impact (i.e., peak linear and rotational accelerations) can be similar between falls and MVA,[5,6] the higher kinetic energy transfer in MVAs increases the likelihood of tissue strain, secondary brain swelling, and concurrent extracranial injuries.[5] Substantial

Table 2 Acute and One-Year Outcome Unweighted and Weighted Descriptives

	Fall			MVA		
	n	Unweighted[a]	Weighted[a]	n	Unweighted[a]	Weighted[a]
Acute outcomes						
GCS eye[b]	2,614	4 [1, 4]	2 [1, 4]	2,497	1 [1, 3]	1 [1, 4]
GCS motor[b]	2,616	6 [3, 6]	5 [1, 6]	2,497	1 [1, 5]	4 [1, 6]
GCS verbal[b]	2,619	4 [1, 5]	2 [1, 4]	2,496	1 [1, 2]	1 [1, 4]
GCS total[b]	2,634	13 [6, 15]	10 [3, 14]	2,503	3 [3, 10]	6 [3, 14]
Days to follow commands	2,637	14.2 (84.0)	13.5 (76.0)	2,482	24.1 (110.1)	20.9 (104.1)
Days in PTA	2,603	172.7 (335.4)	147.3 (309.3)	2,458	151.8 (308.3)	177.1 (333.0)
FIM cognitive admission	2,598	15.7 (7.5)	15.7 (7.8)	2,452	15.2 (7.9)	15.0 (7.6)
FIM cognitive discharge	2,600	23.2 (6.9)	23.8 (6.7)	2,448	23.6 (6.9)	23.5 (6.8)
FIM motor admission	2,171	36.1 (16.5)	36.5 (17.2)	2,112	33.3 (16.6)	32.9 (16.5)
FIM motor discharge	2,177	64.5 (18.4)	66.8 (18.0)	2,110	63.7 (17.5)	62.5 (18.0)
LOS acute	2,671	17.4 (16.7)	19.8 (17.9)	2,510	23.3 (19.3)	23.6 (22.6)
LOS rehab	2,671	23.5 (22.8)	23.4 (23.5)	2,510	25.3 (26.5)	25.1 (24.3)
One-year outcomes						
DRS	2,100	5.4 (8.1)	4.2 (6.9)	1,966	3.3 (4.8)	4.6 (7.7)
PART-O out and about	2,039	1.4 (0.8)	1.4 (0.8)	2,034	1.5 (0.8)	1.6 (0.8)
PART-O productivity	2,039	0.9 (0.9)	1.2 (1.0)	2,034	1.3 (1.0)	1.1 (0.9)
PART-O social	2,039	2.3 (1.0)	2.3 (1.0)	2,034	2.3 (1.0)	2.3 (1.0)
PART-O summary	2,039	1.5 (0.7)	1.6 (0.7)	2,034	1.7 (0.7)	1.7 (0.7)

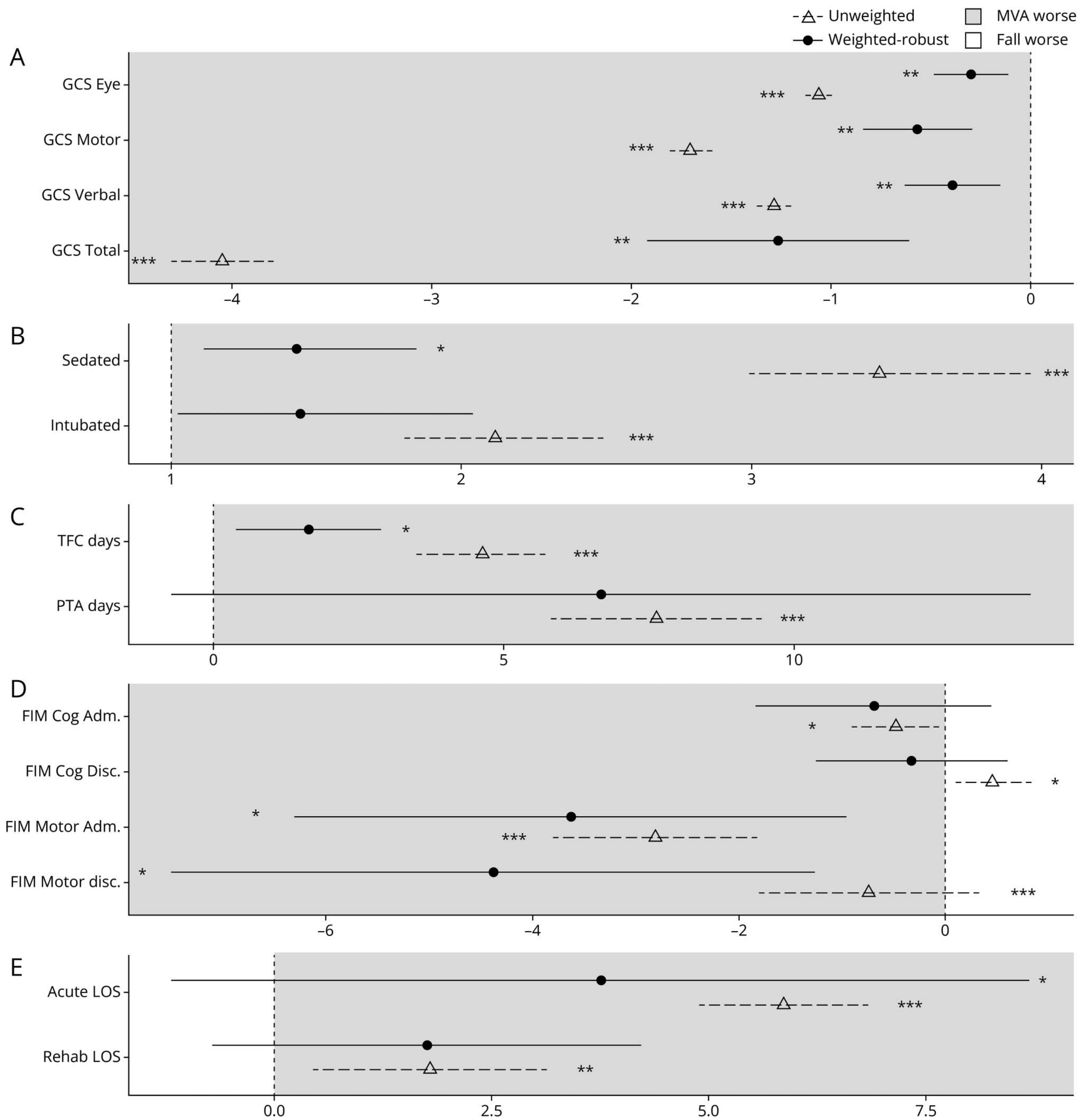
Abbreviations: DRS = Disability Rating Scale; FIM = Functional Independence Measure; GCS = Glasgow Coma Scale; IQR = interquartile range; LOS = length of stay; PART-O = Participation Assessment with Recombined Tools Objective; PTA = post-traumatic amnesia; TBI = traumatic brain injury.
[a] Mean (SD); median [IQR].
[b] GCS scores are reported as median [IQR] due to the ordinal scale but were analyzed as continuous variables to align with previous TBI research.

motor function differences at rehabilitation discharge persisted between MVA-related and fall-related injuries, even after controlling for confounders such as orthopaedic injury, potentially reflecting the residual effects of systemic injury burden. While orthopaedic injuries were identified using ICD codes, residual confounding may exist, because motor deficits could also stem from polytrauma affecting other organ systems, increasing medical complexity and influencing physical function scores (i.e., FIM). Future research should assess polytrauma beyond orthopaedic injury ICD codes to clarify whether MVA-related motor impairments persist. Taken together, these findings emphasize that MOI significantly affects acute injury severity and rehabilitation functional motor outcomes after moderate-severe TBI, even after accounting for systematic differences in demographic and preinjury personal factors.

By 1-year after injury, disability levels did not differ between groups. Despite worse motor function at acute inpatient rehabilitation discharge, the MVA group did not show greater long-term disability. In unweighted models, participants with MVA-related injury had lower levels of disability than those with fall-related injury. However, this difference was no longer present after accounting for demographic and personal factors, including the older age of the fall group, suggesting that long-term disability is better explained by upstream factors than injury mechanism.

The greater acute severity among MVA-related injuries likely reflects higher-velocity impacts and greater overall energy transfer to the body, producing more extensive polytrauma and acute neurologic compromise. However, individuals with MVA-related TBI may experience more complete neurologic recovery once systemic injuries resolve, supported by younger age and fewer comorbidities. Conversely, falls are associated with preinjury medical instability and comorbidities,[11,12,44] which increase with age but are not fully captured by age alone. Those with fall-related TBI may have lower baseline status or less capacity to recover than those injured in MVAs. It is also notable that we did not measure receipt of outpatient rehabilitative therapy after discharge. Previous research suggests that older patients receive less intensive rehabilitation than their younger counterparts,[45,46] which could contribute

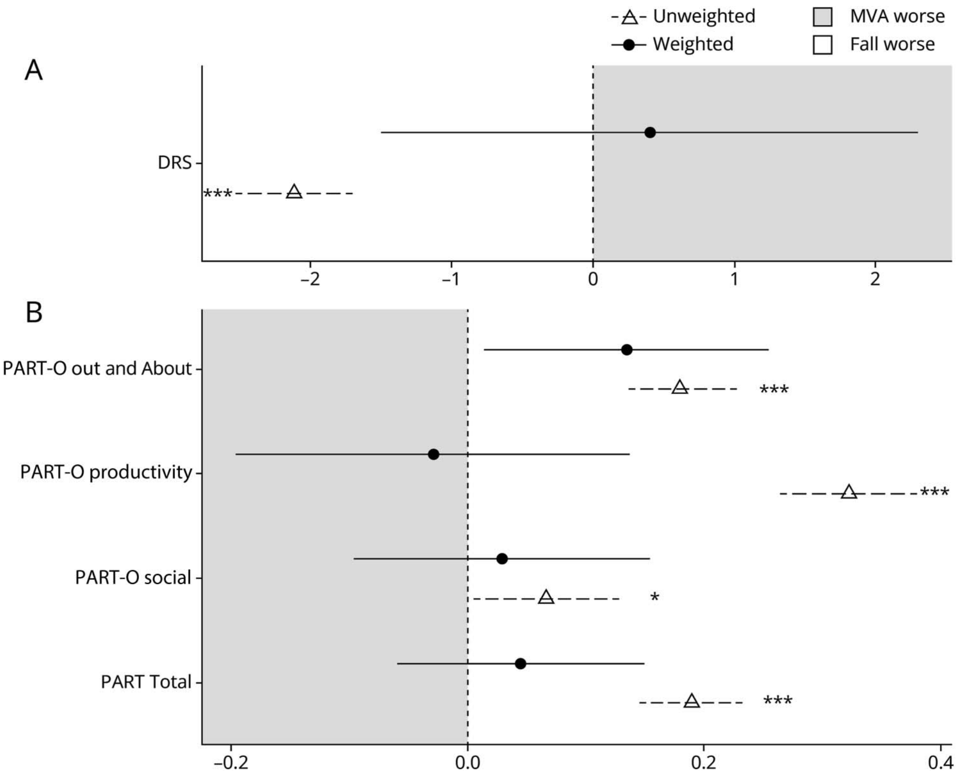
Figure 2 Acute Outcomes



Regression estimates comparing MVA-related TBI with fall-related TBI (reference category) across acute outcomes: (A) GCS subscale and total scores, (B) likelihood of sedation and intubation, (C) TFC and duration of PTA in days, (D) FIM cognitive and motor scores at Adm and Disc, and (E) acute and rehabilitation LOS. Panels A, C, D, and E present the estimated difference in outcomes from linear regression models, while panel B presents the odds ratios estimated from logistic regression models. In most models, applying inverse probability weights to adjust for upstream differences led to attenuated effects, as indicated by estimates approaching the dashed line and signifying smaller differences between groups. Conversely, FIM motor scores at discharge showed a larger difference in weighted models. MVA-related TBI was associated with lower GCS subscale and total scores and lower FIM motor scores, suggesting worse injury severity and functional outcome. MVA-related TBI was associated with a greater likelihood of sedation and intubation and longer TFC, indicating worse injury severity. Estimates falling in the gray area indicate worse outcomes for the MVA-related TBI group. Estimates falling in the white area indicate worse outcomes for the fall-related TBI group. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. For unweighted models, p values were unadjusted; for weighted models, p values were adjusted for multiple testing using the Benjamini-Hochberg method. Adm = admission; Disc = discharge; FIM = Functional Independence Measure; GCS = Glasgow coma scale; LOS = length of stay; MVA = motor vehicle accident; PTA = post-traumatic amnesia; TFC = time to follow commands.

to poorer long-term outcomes in the fall group. Overall, despite worse acute motor outcomes, participants with MVA-related TBI had better recovery when controlling for demographic and personal factors, highlighting the need to consider baseline health, age-related factors, and postacute care access when interpreting long-term outcomes by MOI.

Figure 3 One-Year Outcomes



Linear regression coefficient estimates comparing MVA-related TBI with fall-related TBI (reference category) across one-year outcomes: (A) DRS total score and (B) PART-O subscale and total scores. Applying inverse probability weights to adjust for upstream differences (circle) led to attenuated effects, as indicated by estimates approaching the dashed line and signifying smaller differences between groups. Only PART-O Out and About remained significantly higher in MVA-related TBI compared to fall-related TBI, suggesting greater participation. Estimates falling in the gray area indicate worse outcomes for the MVA-related TBI group. Estimates falling in the white area indicate worse outcomes for the fall-related TBI group. *$p < 0.05$; **$p < 0.01$; ***$p < 0.001$. For unweighted models, p values were unadjusted; for weighted models, p values were adjusted for multiple testing using the Benjamini-Hochberg method. DRS = Disability Rating Scale; MVA = motor vehicle accident; PART-O = Participation Assessment with Recombined Tools Objective; TBI = traumatic brain injury.

Our findings differ from previous literature showing higher mortality and worse global function at 6 months for fall-related injuries compared with other mechanisms, including traffic-related injuries.[7] Although the authors controlled for age, our results suggest that residual differences in other factors may have contributed to these findings. Controlling for age is necessary but not sufficient, as our data indicate that other variables such as medical comorbidity, orthopaedic injury, and education also vary significantly by injury mechanism and should be considered.

Personal and demographic factors may have obscured MOI-related differences in participation outcomes. In unweighted models, participants with MVA had higher PART-O productivity, social relationships, and total scores than those with fall-related TBI, but these differences were no longer significant once upstream differences were controlled. This may be due to systematic differences in age or education between groups as younger individuals or those with higher education may be more likely to be working, attending school, or spending more time taking care of the home and family. After weighting, PART-O out and about scores remained higher in the MVA group compared with the fall group, although this difference was no longer significant after multiple comparison correction. The direction of the effect is consistent with the possibility that greater time since injury and participation in physical and occupational rehabilitation that focuses on improving MVA-related functional deficits could facilitate return to activities outside the home, such as sports or other leisure activities, and may again highlight disparities in rehabilitation care for older adults with TBI,[45,46] but this should be considered exploratory.

Our findings highlight the importance of rigorous confounder adjustment when evaluating outcomes by MOI, as differences between mechanisms, such as MVA and fall, can shift substantially after accounting for upstream personal and health variables. This has clinical implications for prognostication, emphasizing that more severe acute injuries, such as those from MVA, do not necessarily predict greater disability at 1 year after injury. These results reinforce the need for more nuanced approaches to characterizing TBI, moving beyond simplistic severity grading toward multidimensional models that integrate both injury and individual-level modifiers.

This perspective aligns with recent efforts by the NIH National Neurological Disorders and Stroke to refine clinical characterization in TBI, which proposed a framework incorporating 4 pillars—Clinical, Biomarker, Imaging, and Modifier (CBI-M)—to improve precision in TBI classification.[47-49] Although our analysis was not designed explicitly within this framework, it supports the guiding principles of the CBI-M approach. Specifically, our modeling of MOI as an exposure while controlling for demographic and preinjury factors operationalizes key aspects of the *Modifier* pillar, and our use of early trajectory measures such as GCS, PTA, and

TFC reflects proposed markers of acute injury from the *Clinical* pillar. In this sense, our work provides a practical example of how existing longitudinal data can be leveraged to extend CBI-M concepts beyond the acute phase of TBI. Finally, our findings address national priorities highlighted in the recent NASEM workshop on TBI in older adults, which emphasized the need to investigate MOIs beyond falls to improve prevention and rehabilitation strategies.[50]

This study has limitations. The sample included only patients with TBI admitted to inpatient rehabilitation, which limits the generalizability of these findings to the broader TBI population. This restriction excludes individuals with milder injuries who did not require rehabilitation, likely resulting in a fall-related group biased toward more severe TBI cases. Furthermore, the study cohort represents patients with moderate-to-severe TBI who were medically stable enough to participate in rehabilitation, excluding those with more severe injuries who died or required long-term acute care. This selection bias may have influenced outcome differences between mechanisms, making comparisons more conservative. We only examined MVA-related and fall-related TBI, excluding other MOIs such as violence or sport-related injuries, where age and other factors may have different associations with outcomes. Participants aged 80 years and older were excluded to avoid bias from the small number (n = 92) of older adults with MVA-related TBIs. Propensity score weighting can only control for measured covariates, and our models may be influenced by unmeasured confounding from covariates such as day of the week and time of admission, time to admission, or premorbid neurologic decline. In addition, there may be other potential legal, social policy, or environmental issues that affect which individuals present for inpatient rehabilitation. Finally, this study examined outcomes only up to 1 year after injury; therefore, we were unable to determine how MVA-related vs fall-related TBI influences long-term trajectories. Future research should address this important question, carefully accounting for both confounding and also potential nonrandom survival bias in long-term follow-up, particularly among older adults with fall-related TBI.

MOI has unique contributions to predicting acute and 1-year outcomes after TBI. Persons who sustain a TBI from MVA may have worse acute clinical and functional outcomes, but by 1 year after injury, they do not differ in their levels of disability or community participation. These findings highlight the importance of considering the influence of demographic and personal factors when examining the unique contribution of MOI. Furthermore, results suggest that the effect of MOI may differ depending on time since injury (acute vs 1-year after injury) and domain of outcome measured (e.g., cognitive vs motor function). These findings provide novel and important prognostication information for clinicians.

Author Contributions

N.L. de Souza: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. J. Del Pozzo: study concept or design; analysis or interpretation of data. A.-J. Hicks: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data. A. Divecha: analysis or interpretation of data. B. Engelman: analysis or interpretation of data. J. Bogner: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. P.C. Fino: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. E.M. Hade: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. S. Juengst: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. D.W. Klyce: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. P.B. Perrin: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. A. Rabinowitz: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. K. Dams-O'Connor: drafting/revision of the manuscript for content, including medical writing for content; analysis or interpretation of data. R.G. Kumar: drafting/revision of the manuscript for content, including medical writing for content; study concept or design; analysis or interpretation of data.

Study Funding

This work was supported by grants from the National Institute on Disability, Independent Living, and Rehabilitation Research (NIDILRR; under grants 90DPTB0005; 90DPTB0026; 90DPTB0028; 90DPTB0025; 90DPTB0019; 90ARHF0008), grants from NIH Eunice Kennedy Shriver National Institute of Child Health and Human Development (K99HD106060) and the NIH National Institute of Neurological Disorders and Stroke (1RF1NS128961), and in part by an NYU CTSA grant (UL1 TR001445) from the National Center for Advancing Translational Sciences, National Institutes of Health. NIDILRR is a Center within the United States Administration for Community Living (ACL), Department of Health and Human Services (HHS).

Disclosure

The authors report no relevant disclosures. Go to Neurology.org/N for full disclosures.

Publication History

Received by *Neurology*® August 29, 2025. Accepted in final form January 2, 2026. Submitted and externally peer reviewed. The handling editor was Associate Editor Rebecca Burch, MD.

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